AI Assistant Coding

Assignment – 14.1

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Batch : 08

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Task 1 – Portfolio Website Design

You are building a personal portfolio website to showcase your work.

Requirements:

• Create sections for About Me, Projects, and Contact.

• Use AI to:

o Suggest color palettes and typography.

o Create a responsive layout with Grid/Flexbox.

o Add smooth scrolling navigation.

Task 1 – Portfolio Website Design

```
from IPython.display import display, HTML

html = """
<!DOCTYPE html>
<html>
<head>

<style>

body{
font-family: Arial;
margin:0;
scroll-behavior:smooth;
}

nav{
background:#1e3a8a;
padding:15px;
position:fixed;
width:100%;
}

nav a{
color:white;
margin:15px;
```

```
margin:15px;
text-decoration:none;
font-weight:bold;
}

section{
padding:100px 40px;
height:100vh;
}

.projects{
display:grid;
grid-template-columns:repeat(auto-fit,minmax(200px,1fr));
gap:20px;
}

.card{
background:#f4f4f4;
padding:20px;
border-radius:8px;
}

</style>
</head>

<body>

<nav>
<a href="#about">About</a>
<a href="#projects">Projects</a>
```

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<a href="#about">About</a>
<a href="#projects">Projects</a>
<a href="#contact">Contact</a>
</nav>

<section id="about">
<h2>About Me</h2>
<p>I am a web developer building modern websites.</p>
</section>

<section id="projects">
<h2>Projects</h2>

<div class="projects">
<div class="card">Project 1</div>
<div class="card">Project 2</div>
<div class="card">Project 3</div>
</div>

</section>

<section id="contact">
<h2>Contact</h2>
<p>Email : example@gmail.com</p>
</section>

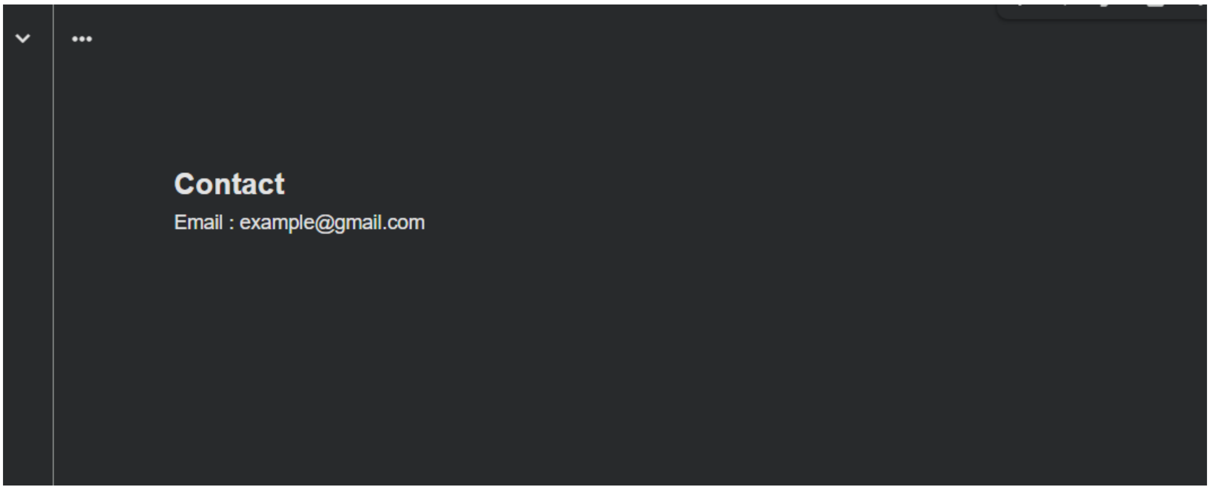
</body>
</html>
"""
```

```
display(HTML(html))
```

About Projects Contact

About Me

I am a web developer building modern websites.



Observation:

The portfolio website was successfully created with sections for About Me, Projects, and Contact. CSS Grid was used to organize the project cards in a responsive layout that adjusts automatically to different screen sizes. Smooth scrolling navigation allowed easy movement between sections. The chosen color palette and typography improved readability and gave the website a professional appearance.

Task 2 – Online Store Product Page

Design a product display page for an online store.

Requirements:

• Display product image, title, price, and "Add to Cart" button.

• Use AI to:

o Style with BEM methodology.

o Make layout responsive.

o Add hover effects and "Add to Cart" alert.

Task 2 – Online Store Product Page

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```
from IPython.display import display, HTML

html = """

<style>

.product{
text-align:center;
font-family:Arial;
max-width:300px;
margin:auto;
}

.product__img{
width:100%;
}

.product__btn{
background:#28a745;
color:white;
padding:10px 20px;
border:none;
cursor:pointer;
}

.product__btn:hover{
background:#1e7e34;
```

```
background:#1e7e34;
}

</style>

<script>

function addCart(){
alert("Product added to cart!")
}

</script>

<div class="product">

<img class="product__img"
src="https://via.placeholder.com/300">

<h2 class="product__title">Smart Watch</h2>

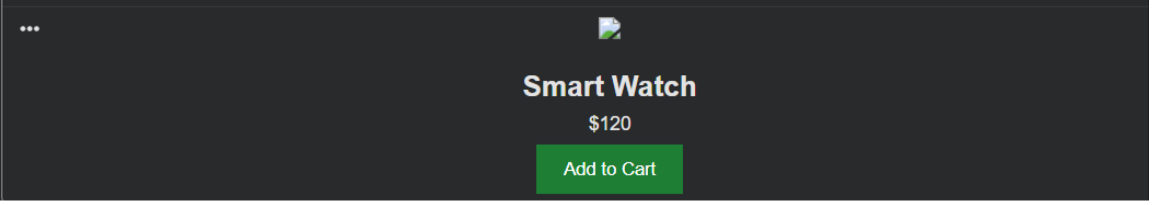
<p class="product__price">$120</p>

<button class="product__btn"
onclick="addCart()">Add to Cart</button>

</div>
"""

display(HTML(html))
```

```
display(HTML(html))
```



Observation:

The product page displayed the product image, title, price, and an "Add to Cart" button clearly. BEM methodology was used for structuring CSS classes, making the design modular and easier to maintain. Hover effects improved the visual interaction with the button, and clicking the "Add to Cart" button successfully triggered a JavaScript alert confirming the action.

Task 3 – Event Registration Form

Build an event registration form for a conference.

Requirements:

- Collect name, email, phone number, and session selection.
- Use AI to:

o Add form validation with JavaScript.

o Make the form accessible with labels and ARIA.

o Style with a professional look.

Task 3 – Event Registration Form

```
from IPython.display import display, HTML

html = """

<style>

form{
width:300px;
margin:auto;
font-family:Arial;
}

input,select{
width:100%;
padding:10px;
margin:10px 0;
}

button{
background:#007bff;
color:white;
padding:10px;
border:none;
}

</style>
```

```
<script>

function validateForm(){

let name=document.forms["form"]["name"].value

if(name==""){
alert("Name must be filled")
return false
}

return true
}

</script>

<h2 align=center>Event Registration</h2>

<form name="form" onsubmit="return validateForm()">

<label>Name</label>
<input type="text" name="name">

<label>Email</label>
<input type="email">

<label>Phone</label>
<input type="tel">
```

```
<input type="tel">

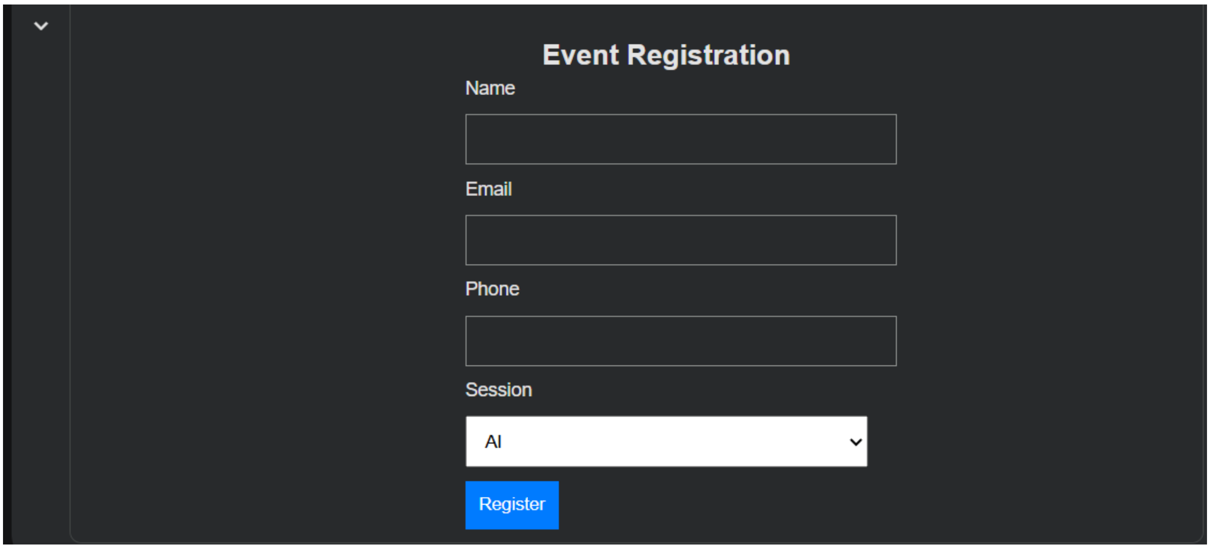
<label>Session</label>

<select>
<option>AI</option>
<option>Web Development</option>
<option>Cyber Security</option>
</select>

<button type="submit">Register</button>

</form>
"""

display(HTML(html))
```



Observation:

The event registration form collected user details such as name, email, phone number, and session selection. JavaScript form validation ensured that required fields were filled before submission. Proper labels were used to make the form accessible and easier to understand. The styling provided a clean and professional interface for user interaction.

Task 4 – Resume Website

Build a one-page online resume.

Requirements:

• Sections: Profile Summary, Skills, Education, Experience.

• Use AI to:

o Propose professional typography styles.

o Align sections using CSS Grid.

o Add smooth scroll navigation between sections.

Task 4 – Resume Website

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```
from IPython.display import display, HTML

html = """

<style>

body{
font-family:Georgia;
scroll-behavior:smooth;
}

nav{
background:#333;
padding:10px;
}

nav a{
color:white;
margin:10px;
text-decoration:none;
}

.container{
display:grid;
grid-template-columns:1fr;
gap:30px;
padding:40px;
```

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```
padding:40px;
}

section{
border-bottom:1px solid #ddd;
padding:20px;
}

</style>

<nav>

<a href="#profile">Profile</a>
<a href="#skills">Skills</a>
<a href="#education">Education</a>
<a href="#experience">Experience</a>

</nav>

<div class="container">

<section id="profile">
<h2>Profile Summary</h2>
<p>Motivated software developer.</p>
</section>

<section id="skills">
<h2>Skills</h2>
<p>Python, HTML, CSS, JavaScript</p>
</section>
```

```
<p>Python, HTML, CSS, JavaScript</p>
</section>

<section id="education">
<h2>Education</h2>
<p>B.Tech Computer Science</p>
</section>

<section id="experience">
<h2>Experience</h2>
<p>Web Developer Intern</p>
</section>

</div>
"""

display(HTML(html))
```

Profile Skills Education Experience

Profile Summary

Motivated software developer.

Skills

Python, HTML, CSS, JavaScript

Education

B.Tech Computer Science

Skills

Python, HTML, CSS, JavaScript

Education

B.Tech Computer Science

Experience

Web Developer Intern

Observation:

A one-page resume website was created containing sections for Profile Summary, Skills, Education, and Experience. CSS Grid was used to align the sections neatly, improving layout organization. Smooth scrolling navigation allowed users to jump quickly between sections. The typography styles made the content readable and gave the resume a professional presentation.

Task 5 – Restaurant Menu Page

Design a digital menu page for a restaurant.

Requirements:

• Display menu items with name, description, and price.

• Use AI to:

o Suggest a food-friendly color palette.

o Design a responsive card-based layout.

o Add hover effects on menu items.

Task 5 – Restaurant Menu Page

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```
from IPython.display import display, HTML

html = """

<style>

body{
font-family:Arial;
background:#fff8f0;
}

.menu{
display:grid;
grid-template-columns:repeat(auto-fit,minmax(200px,1fr));
gap:20px;
padding:40px;
}

.card{
background:white;
padding:20px;
border-radius:10px;
box-shadow:0 3px 6px rgba(0,0,0,0.2);
transition:0.3s;
}
```

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```
}

.card:hover{
transform:scale(1.05);
}

.price{
color:#e63946;
font-weight:bold;
}

</style>

<h1 align=center>Restaurant Menu</h1>

<div class="menu">

<div class="card">
<h3>Pizza</h3>
<p>Cheese pizza</p>
<p class="price">$10</p>
</div>

<div class="card">
<h3>Burger</h3>
<p>Chicken burger</p>
<p class="price">$8</p>
</div>
```

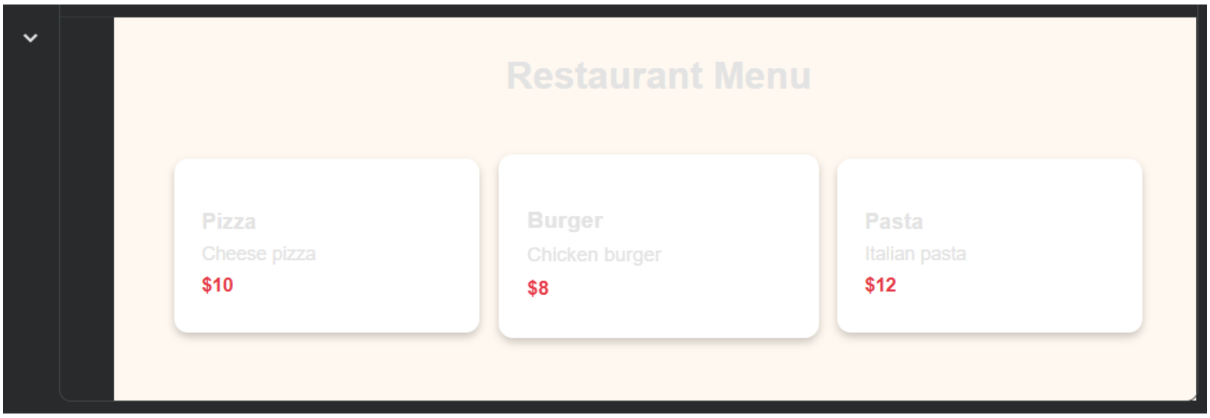
```
<h3>Burger</h3>
<p>Chicken burger</p>
<p class="price">$8</p>
</div>

<div class="card">
<h3>Pasta</h3>
<p>Italian pasta</p>
<p class="price">$12</p>
</div>

</div>

"""

display(HTML(html))
```



Observation:

The restaurant menu page displayed menu items using a responsive card-based layout. Each card included the item name, description, and price. A food-friendly color palette was applied to create an attractive visual theme. Hover effects on the menu cards enhanced user interaction and made the menu more engaging.